

Safety & Behaviour Rules

In the Cleanrooms of Else Kooi Laboratory

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Dear user of the EKL facilities

I cordially welcome you to the Else Kooi Laboratories. We, and our labs, are there to help to realise your ideas in silicon.

The centre houses a number of Cleanrooms and Measurement and Characterisation labs. A lot of gases and chemicals are used that need special handling and can pose a potential hazard if handled improperly.

In such an environment it is paramount that there are strict rules about safety and working conditions and that the users have sufficient knowledge on how to behave and act in our facilities.

To guarantee this safety, the enclosed safety and behaviour rules have been formulated and handed to you.

I want to assure that everybody who is working within EKL is acquainted with these rules: for your own safety, the safety of your colleagues, and the preservation of the most optimal research climate.

You **MUST** read this whole booklet and you can hold on to this booklet for future reference.

If you have any questions, remarks or suggestions after reading this, you are welcome to contact me or the Safety Committee.

The basic rules for safety questions are threefold:

1. Safety goes first.
2. There are no stupid questions.
3. If you hesitate, please ask.

Notwithstanding the risks intrinsic to working with chemicals, gases and process equipment, as present within EKL, no calamity has occurred since the foundation in 1987. Our intention is to keep it this way!

Furthermore our goal is to reduce the occurrences of small incidents to a minimum. In this we can only succeed if everybody shares this commitment. While working here, you are part of us: it is your cleanroom too.

EKL is a dynamic institute. Therefore the safety and behaviour rules are subject to change. Alterations are announced through an e-mail to the users of EKL and on the EKL website.

We also ask you to do a yearly exam to keep your knowledge on safety at the highest possible level during your stay in EKL.

I wish you all the best in your work and research within EKL.

Yours sincerely,

Casper Juffermans
Director Else Kooi Laboraories

Summary

Safety training

All people that start working in the laboratories, whether they are postdoc, student, operator or cleaner, first need to take a safety training. This training comprises a theoretical and a practical part.

Rules

A set of safety and behaviour rules has been formulated.

It is handed to all Cleanroom workers and forms the base guide for clean and safe work.

Experienced operators

The back bone for safety is a team of experienced operators, supported by an alert and devoted facility group.

About half of this team is member of the Emergency Response Team.

They can tackle incidents ranging from personal injuries and contaminations, to putting out a fire and rescuing people. Together with the fire department of Delft, training for the ERT teamare is organised at least 8 times per year with regular intervals.

Emergency buttons

To communicate incidents there is the aid of emergency circuits and buttons. A dense array of fire, smoke, gas and water sensors is installed, along with manual alarm devices. Any alarm is immediately communicated to a telephony centre, that pages the alarm to the beepers of the Emergency Response Team and, if needed, to the fire department of Delft.

Safety management

The final responsibility for safety lies with the EKLC director. About safety issues, he is advised by the Safety Committee that consists of process technologists with different expertise, the commander of the Emergency Response Team and Occupational Health Service officer. Together they have a helicopter view on the EKL laboratories. A meeting of this safety committee is organised every 6 weeks.

Facility group

The maintenance of the lab facilities, like air flow, gas lines, gas supply, chemical supply, cooling water etc are in the capable hands of the facility group. Twice a year the lab will be down for about one week for maintenance, usually in April and October. For questions about these general facilities you can contact facility team (see contact list on last page).

Also the lab is kept clean by people with a special training to clean Cleanrooms.

Ordering of new chemicals

You are not allowed to bring your own chemicals into the labs. All chemicals have to be ordered through the facility group. Before ordering you need permission for use (contact Robert Verhoeven, r.c.m.verhoeven@tudelft.nl and show a proper Safety Data Sheet (SDS).



*I forgot to look at the clock
and now I'm stuck in the
MEMSlab!*

I was so busy with my work in the MEMSlab, that I did not look at the clock. It was 18:15 hrs and I could not get out of the lab anymore. What could I do?

Nobody answered the phone, everyone was gone home and I was all alone.

Luckily I kept calm and I decided to break the glass of the green box at the door to open it. Then I sent a mail to the facility people to tell them that I was the one who broke the door seal. The next morning I went by their office to apologize. Meanwhile the facility people did reset the door lock

They told me I offended 2 rules:

- If you work in the MEMSlab there should always be a second person with entrance rights on call in the EKL building.
- You should get out of the lab before 18:00hrs otherwise the door is locked.

But they complimented me with the fact that I handled the situation properly:

Break the glass

Send an email

Visit the facility people in person

Laboratories

EKL has a [ISO 5 \(Class 100\)](#) and a [ISO 7 \(Class 10000\)](#) Cleanroom. The [Special Applications Lab \(SALab\)](#) is part of the Class 100 area.

In these areas the temperature is kept constant to 20 ± 0.2 °C and the humidity is about $48\% \pm 5\%$. The [Pavilion/MEMSlab](#) is locally [ISO 5 \(Class 100\)](#).

The [Chemical Room](#) is used for transfer of goods between outside and the Cleanroom and to clean vacuum parts.

Other areas are the [Hoist Area](#) where liquid nitrogen can be transferred to Dewar vessels and the [Basement](#) where all the vacuum pumps are located.

Cleanroom airflow conditions

[Pavilion/MEMSlab](#)

Two states are distinguished:

1. Monday until Friday from 07.30 till 18.00: High air inlet and extraction: conditions meet the requirements and work can be carried out safely.
2. Monday until Friday from 18.00 till 07.30 and on Saturdays, Sundays and during holidays:
 - Low air inlet and extraction
 - **Conditions do NOT meet the requirements for safe working.**
 - For all gas boxes and chemical cabinets the air extraction is always high

Consequences:

The MEMSlab is closed in the weekends and on workdays from 18:00-8:00 hrs.

Only with special permission it is possible to switch the air to high inlet condition for a limited time till max 21:00 hrs and only from Monday till Friday.

[Class 100 and Class 10000 cleanrooms](#)

The conditions in the Class 100 and Class 10000 cleanrooms always meet the specifications for safe working. Outside office times the humidity is not regulated, which could influence critical lithographic processes.

[Maintenance week](#)

During the maintenance weeks, usually in April and in October, it is not allowed to work in the Cleanrooms or the MEMSlab. There could be interruptions in airflow, in exhaust, in gas/water/electricity. Also the EKL staff will do maintenance on the equipment.

Safe working conditions do not exist.

Access

Access to the Cleanrooms is granted after completion of the safety course.

For the CR100 extra training is necessary, ranging from a module/equipment training to a full process training. For the other areas only equipment training is necessary.

1. The safety and behaviour rules must be known by any means. Changes will be communicated through e-mail to all the users of the Else Kooi Laboratories and on the web: <http://www.ekl.tudelft.nl/EKLi/IntranetHome.php> and during 2 weeks on the publication boards within EKL.
2. An evacuation plan is located at many places in the building. On this plan the emergency exits are indicated as well as instructions how to act in case of an alarm.
3. It is only allowed to access the cleanroom with a validated campus card. This card is personal and only gives access to rooms and equipment to which you are authorized. For short term visitors and users special guest card are issued. This card is personalised for the time of use.
4. It is not allowed to accompany anybody, or granting access to others not holding a validated card. The fire department of Delft requires unambiguous registration and monitoring 24 hours a day of the presence of people for all relevant areas via the access control system.
5. It is not allowed to lend your card to somebody else. In case you have lost your card, immediately inform the secretary so she can inactivate the card code from the access control system.
6. New users have a so called 'pupil status' and are only allowed to work under supervision of a mentor. They only can enter and leave the cleanroom from Monday to Friday between 08.00 and 18.00.
After passing a test successfully, the cleanroom user can be authorized to work independently on specific equipment. BSc and MSc students will always have a 'pupil' status.
7. Access only when proper clothing is used in the correct way (chapter: Clothing).
8. Doors of the chemical room and the chemical cabinets must be closed at all times.
The door between the chemical room and the hoist area may only be used for changing chemicals and introducing equipment in the CR 10000. The chemical room is also part of the Cleanroom, therefore minimise the presence in this area without wearing proper clothing. Place materials directly behind the door opening, but keep the passage clear. Close the door as soon as possible, and enter the Cleanroom via the changing room for further transport of the materials.

I want my neighbour to show my work in the lab ...

My neighbour is really interested in my work in the lab, so I invited him into the lab to see what nice things I'm making.

You need SPECIAL PERMISSION from the EKL director to let visitors enter the Cleanrooms.

With this permission you go to the secretary to ask for a visitor badge that gives temporary access rights. The host must have the proper training to enter the lab

Permission to let your friends or family into the labs is NOT granted. Only work related guests and special groups are welcome.

Working during off-hours

Working in the Cleanrooms is under certain limitations from [Monday until Friday between 17:00 and 07:30](#) and on [Saturdays, Sundays and during holidays](#). During those hours no staff or Emergency Response Team members are on site.

Depending on the situation and the equipment to be used the following rules apply:

1. If you are not working with chemicals you can work on your own. However no maintenance on equipment is allowed.

2. It is never allowed to work with chemicals on your own. So no mixing, no chemicals in beakers etc. Also no use of standard - prefilled- baths.

Working with chemicals requires a buddy. **Also during office hours.**

3. In the MEMSlab working with chemicals during staff hours (7:30-17:00) is allowed when a buddy is present. Outside office hours working with chemicals is ALWAYS forbidden.

The Buddy System

As described above the buddy system means that there is always a second person present. The buddy must be in the same working area but not subject to the same dangers and must be qualified for working in the Cleanroom/MEMSlab.

In case of trainees, approval by the mentor to work with a buddy must be given.

My KOH process will be ready at 23:00 hrs...

KOH etching is a chemical process. So you are not allowed to take your wafer out without a buddy.

So you need to bring in a colleague WITH THE SAME RIGHTS AND EXPERIENCE into the lab.

Or you had to either start earlier in the day, or take your wafer out at 17:00 hrs and go on the next morning.

Note: if you process in the MEMSlab, you cannot do this, even with a buddy, because after 17:00 hrs it is forbidden to work with chemicals.

Note that KOH is highly corrosive and toxic.

Clothing

- You are obliged to wear the prescribed clothing. When entering the Cleanroom for the first time you will get instructions how to wear the Cleanroom clothes.
- Regular staff wear blue clothes. Students without a diploma and visitors wear white clothes.
- Upon leaving, you should hang the clothes back in the coat-rack and put back the overshoes in the proper location.
- Dirty cleanroom garment has to be put in the laundry basket. The baskets are emptied every Tuesday. Clean outfits are available from the stock cabinet. Change your Cleanroom suit at least once a week, preferably on Tuesdays. Keep the changing room tidy.
- Wear trousers and closed shoes under your cleanroom clothing, also in the summertime. Cleanroom suits do not protect you against chemicals, they only protect the environment against the dust you produce.

See the movie about gowning on the EKL intranet: <http://ekl.tudelft.nl/EKLi/Safety.php>



Video Surveillance

The EKL building and the laboratories are equipped with a CCTV system.

Cameras are positioned at the entrances of EKL, the basement and near all the wetbenches in the laboratoria.

The images can only be viewed by the director or his assigned delegate. Images will be saved a maximum of 1 week.

The images will only be used to review what happened during an incident, if neccessary.

Working with chemicals

General

- Before application, consult the Material Safety Data Sheets (MSDS) of the chemicals you are going to use.
- You are allowed to work with chemicals after you have passed the course “[Introduction in the EKL Cleanrooms](#)” and after permission of the responsible persons of the ICP, PVMD or Mapper group.
Working with chemicals in the MEMS laboratory is only allowed if regular telephonic contact is made with a second person present in EKL.
- Only the EKL director or its designated delegates can give approval for the introduction of other chemicals than the ones already available. All orders must be done by the facility group.
- It is not allowed to bring your own chemicals. Only in very special situations the management can decide to deviate from this rule. The facility group must be supplied with relevant data such as Material Safety Data Sheets (MSDS).
- The facility group is not allowed to order other than standard chemicals without permission of the EKL director or its designated delegates.

Handling chemicals

- You are obliged to wear the appropriate personal protection.
- Use the right fumehood, do not use a flow bench for handling chemicals.
- Avoid inhalation of and exposure to vapours. Never work with your head inside the laminar flow of a clean bench or fume-cupboard.
- Never leave residues of chemicals when you are ready with your activities. Clean up everything and leave a clean workplace behind, to avoid risk of injury to your colleagues.
- Wash your hands thoroughly using soap upon completing your activities.
- Never place organic solvents close to or directly on hot plates to prevent fire risk. Organic solvents may only be heated in a “bain-marie”.
- When preparing an aqueous solution, always add the acid or alkaline to the water.
so: ALWAYS WATER FIRST!
- Keep organic liquids strictly separated from acids, peroxides, alkalines, bromides, etc. because of explosion-risk. (E.g. do not use Acetone and 100% HNO_3 in the same fumehood).

Chemical Waste Disposal

Small Chemical Waste (KCA) is transported according to the statutory regulations:

- Organic liquids in the appropriate waste jerrycans. There are different cans for halogen-free and halogen containing organics.
- Inorganic waste (acids, alkalines) with excess of water through the drain of a clean bench or using the Venturi vacuum pump to the neutralisation tank.
Never pour concentrated acids or bases down the drain, always dilute with lots of water. Otherwise the piping will deteriorate and the neutralisation tank will give an alarm because it cannot handle the high/low pH values.
- For wet etchants for metals, III-V compounds and other environmental polluting chemicals special rules apply. There are special waste containers for this.
- When transporting chemicals over a longer distance (e.g. from the chemical room to a wet bench), always use a transport cart or crate. For shorter distances always hold the bottle with both hands. Put empty bottles to the designated places, do not place them back in the cabinet. Fresh chemical solutions to be stored should always be labelled with: name(s) of the chemical(s), owner, telephone number and date of preparation.
- Never pour excess chemicals back into the original bottle. You will contaminate the bottle and its contents.

Chemical Spills and other calamities

- Spilling of chemicals must always be reported to the Emergency Response Team by:
 1. Pressing the yellow calamity button in case of direct danger for the victim and / or bystander(s).
 2. Reporting by phone: see the phone list next to the phone.
- When chemicals are spilt in the open area of the workplace, in case of acids and alkalines: press the CALAMITY button.
Organic liquids: quickly absorb the liquid with tissues and put these in the laminar flow of the clean bench. Avoid contact with hot baking plates.
- How to handle in case of incidents with chemicals and gases will be explained in the chapter: How to act in case of accidents with chemicals or gases.

Pregnancy

In case of pregnancy or breast feeding working with chemicals should be limited.

The National Institute for Public Health and the Environment (RIVM) advises not to work with open (organic) chemicals eg photoresists.

Since blood between mother and foetus is shared avoid absorption of chemicals through skin or otherwise. Especially the first stages of pregnancy are critical in this respect.

Also males in the child planning stage should be careful.

EKL advises not to work in the lithographic areas and not to work with chemicals.

Working against the advice of EKL is on the users own risk.

Look on the chemical container for the H-numbers. They give information about the harmfulness of the chemical. For pregnancy the following H-numbers are important:

H360 – May damage fertility or the unborn child

H361 – Suspected of damaging fertility or the unborn child

H362 – May cause harm to breast-fed children.

In the old coding system the following R numbers apply: R60, R61, R63 and R64.

A selection of potential harmful substances that are/has been used at EKL. A more complete list can be found on the EKL intranet.

Aluminium compounds that are soluble

Chloroform, Formaldehyde

Chromium(VI)compounds (not allowed anymore at EKL)

Cyclohexanol

1,2 diethoxyethane

Ethanol

Hexane

Potassiumdichromate

Coppersulfate

Methanol

Nickel compounds

NMP

Photoresists

Toluene

TCA

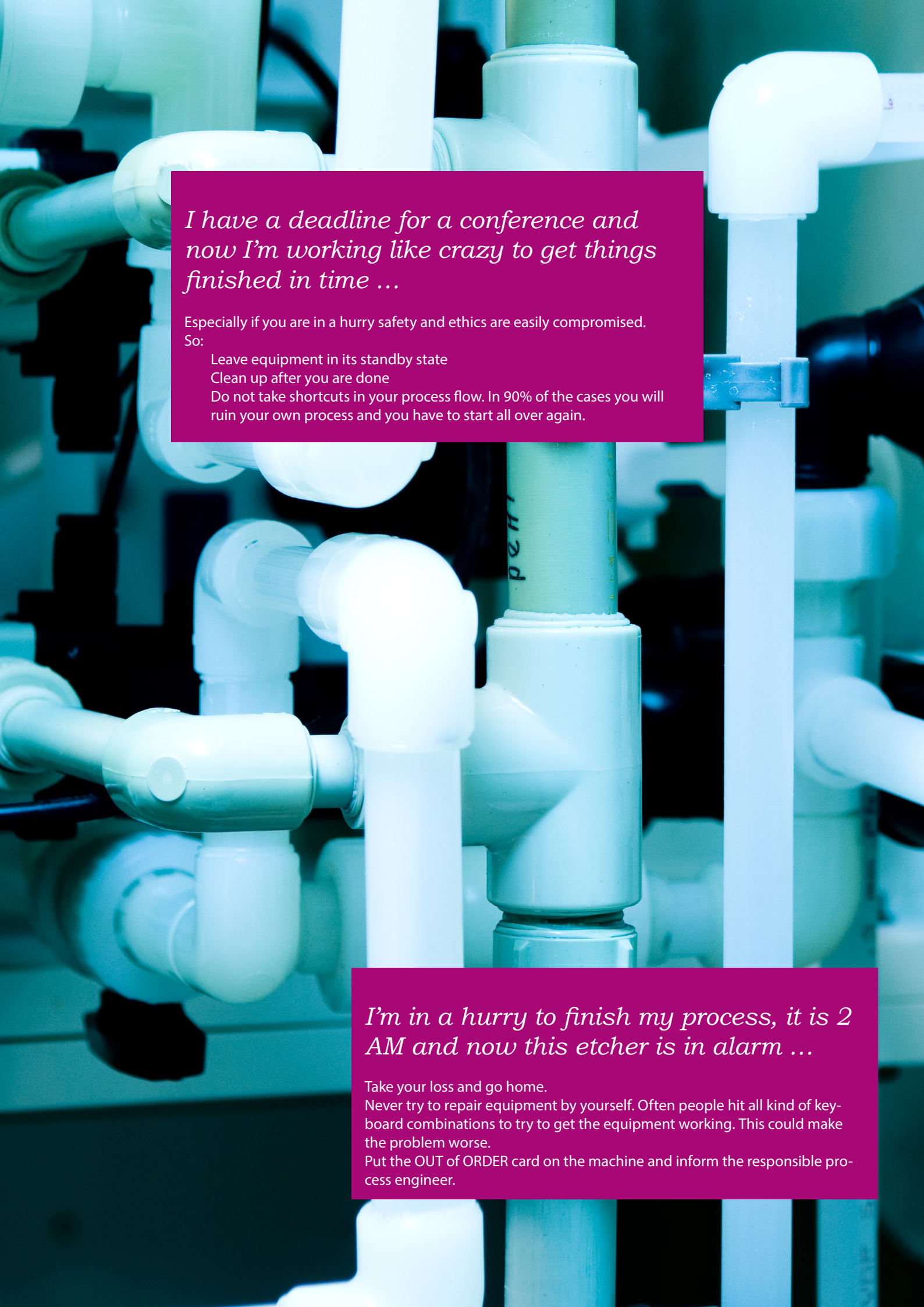
Xylene

You can still work with most of these chemicals in fume hoods. Photoresist coating on the EVG coat-er is no problem.

Spray coating or manual coating is not advised.

Also do not use organic solvents, like IPA, Methanol, Acetone etc.

For more information contact the Safety Committee.



I have a deadline for a conference and now I'm working like crazy to get things finished in time ...

Especially if you are in a hurry safety and ethics are easily compromised.
So:

Leave equipment in its standby state

Clean up after you are done

Do not take shortcuts in your process flow. In 90% of the cases you will ruin your own process and you have to start all over again.

I'm in a hurry to finish my process, it is 2 AM and now this etcher is in alarm ...

Take your loss and go home.

Never try to repair equipment by yourself. Often people hit all kind of keyboard combinations to try to get the equipment working. This could make the problem worse.

Put the OUT of ORDER card on the machine and inform the responsible process engineer.

Working with equipment

New Cleanroom users exclusively work under supervision of a mentor. Independent operation of equipment is allowed after the new user has passed the appropriate test.

- When a machine goes into failure, immediately warn the responsible person or his deputy. It is not allowed to do any maintenance yourself.
- Before starting your activities, convince yourself that the machine works properly and is in a good condition. After you have used the machine leave it behind in its original state.
- Never change the pressure settings on the gas panels that supply the machines. This may affect proper performance of the machine, possibly damage it and even lead to hazardous situations.
- When a machine does not work properly or may cause danger, push the emergency stop button of the machine. If the problem is not solved or the danger not cleared, then push the closest red EMERGENCY STOP button. Refer to the chapter: Emergency procedures.
- When big problems occur like leakage of gasses, fumes and / or liquids, immediately push the closest red EMERGENCY STOP button. Refer to the chapter: Emergency procedures.

Prevention of contamination: CR behaviour

- Wear proper clothing in a correct way. Refer to the chapter: Clothing
- Use suitable gloves for critical work: amongst others in the clean benches, during all litho-experiments and during loading/unloading vacuum equipment.
- Only use the special Cleanroom paper for making notes. Normal paper is only allowed when sealed in plastic. Wood, cardboard, packing materials etc. are strictly forbidden.
- Limit the number of things to be taken with you to the minimum: e.g. substrate boxes, a pair of tweezers, resist syringe, ball pen or felt pen (no pencil!)
- Redundant clothes (jacket, jumper, etc.) and bags should be put in the wardrobe. Valuables can be stored in lockers in the clothing room.
- Avoid turbulence as much as possible: do not hurry, minimise the used working area in a clean bench, but keep your work in the centre of the down flow. Be aware of the dust cone around you.

Ethics

- In case you make an error: you **MUST** always report this to a staff member.
E.g. contamination of a chemical bath that is also used by others, spilling of chemicals etc. Everybody makes mistakes, but **not telling it is considered as a severe lack of responsibility**, that can ruin other people's process or worse: other people's safety.
- Cleanup your workplace after you are done, bring equipment back in its standby state.
- It is the responsibility of all lab users, students and staff, to correct improper behaviour of others.
But do this in a friendly way. If you run into problems contact a staff member that you think you can trust or contact the Safety Committee.
- Not following the rules can result in revoking your access privileges or other disciplinary actions.

Emergency procedures

There are different ways to call for help:

Calamity button (yellow button)

- If an unknown liquid is on the floor, in case of chemical spills
- If someone is exposed to chemical liquids
- If someone is exposed to chemical fumes and/or gases.

Result

- Dependent on the location where the button is pressed on the warning signs, "cleanroom verlaten" (leave the Cleanroom), "paviljoen verlaten" (leave the pavilion) or "kelder verlaten" (leave the basement) light up and an acoustic signal sounds
- The Emergency Response Team is alarmed.
- **Processes do NOT stop, supply of process gases is continued. Cleanroom airflow is put on half power.**

Emergency stop button (red button)

- When you suspect that a machine is not working properly and possess a threat to people present in the Cleanroom
- When gases leak somewhere.

Result

- On the warning signs, "cleanroom verlaten", "paviljoen verlaten" or "kelder verlaten" light up and an acoustic signal sounds.
- The Emergency Response Team is alarmed.
- **All processes in the CR 100, CR 10000 as well as in the pavilion will stop, supply of process gases is **stopped**, all lids of the chemical bath are closed. Cleanroom airflow is put on half power.**

Fire alarm (red fire-alarm box)

- Push in case of fire.

Result

- On the warning signs, "cleanroom verlaten", "paviljoen verlaten" or "kelder verlaten" light up and an acoustic signal sounds.
- The Emergency Response Team and the fire department of Delft are alarmed.
- **All processes in the CR 100, CR 10000 as well as in the pavilion/MEMSlab will **stop**, supply of process gases is halted. Cleanroom airflow is put on half power.**

Phone

- Sometimes there is an emergency that is not important enough that you have to press a button (eg very local or small contamination) in that case you can contact 24/7 the facility person on duty by calling **015- 2510192**. Only call this number in real emergencies, so not to get supplies.
- In case of fire and/or an emergency, in addition to the alarm buttons described above, one can also give the alarm by dialling **112** without the 0 in front from a fixed phone. You will be connected to the university emergency centre, and they will alarm the emergency services.
- If you dial from a mobile phone 112 will connect you to the national alarm centre. This is not preferred. From a mobile phone dial **015-2781226** to connect you to the university emergency centre.

Automatic Alarms

Two alarms are generated automatically: the GAS ALARM and the AUTOMATIC FIRE ALARM.

GAS ALARM

- This alarm is generated automatically when an excessive amount of gas is detected in a given area.

Result

- On the warning signs, "gas alarm cleanroom verlaten", "gas alarm paviljoen verlaten" or "gas alarm kelder verlaten" light up and an acoustic signal sounds.
- The Emergency Response Team and the fire department of Delft are alarmed.
- All processes in the CR 100, CR 10000 as well as in the pavilion will **stop**, supply of process gases is halted. Cleanroom airflow is put on half power.

AUTOMATIC FIRE ALARM

This alarm is generated if a automatic fire alarm detects an excessive amount of smoke or heat in an area.

Result

- On the warning signs, "cleanroom verlaten", "paviljoen verlaten" or "kelder verlaten" light up and an acoustic signal sounds.
- The Dimes Emergency Response Team and the fire department of Delft are alarmed.
- All processes in the CR 100, CR 10000 as well as in the pavilion will **stop**, supply of process gases is halted. Cleanroom airflow is put on half power.

Alarm signals

- As soon as the acoustic and visual signals sound and light up respectively, you must leave the Cleanroom, Pavilion/Memslab or basement immediately.
- Do not waste time getting out of your Cleanroom suit, just walk out. If necessary, use the nearest emergency exit.
- You must report yourself at the meeting point (the central hall of EKL) at room 001 immediately, log out with your personal key card at the main entrance. This is to make sure the access system knows that you are no longer in the lab. Wait for instructions from the Emergency Response Team.
- If there is an alarm outside office times, DO NOT go home, but wait for the ERT or the Firebrigade and explain what happened. (It can take 20 min before the ERT comes, because they are alarmed at home and have to come to EKL.)

Escape routes

- **CR 100 and CR 10000**
In the Cleanroom the emergency exits are located in the 4 corners.
- **Pavilion/MEMSlab**
In the pavilion/MEMSlab there are emergency exits in every bay.
- **Basement**
In the basements there are 3 emergency exits.
- After you have left the area you must report yourself immediately at the meeting point, log out with your personal key card at the main entrance. Wait for instructions from the Emergency Response Team.
- All doors are unlocked if an alarm goes off. If a door can not be opened in an emergency situation, you can still open it by pushing the glass window of the **green box** besides the door. This option should only be used in alarm situations.
- If an emergency situation happens after normal working hours, press the appropriate alarm buttons and dial 112 from a fixed phone. Do not go home but wait at the meeting point for the Emergency Response Team to come (from home). Then tell them what happened.



The wafer carrier fell into the HF bath and HF splashed in my face ...

- Ask for help!
- Let someone push the Yellow CALAMITY button to get help ASAP
- Take off your safety goggles and your cleanroom clothes.
- Spray with Hexafluorine on the parts of your body that are contaminated, also your eyes if necessary.
- Make sure other people do not get contaminated also.
- Do not panic

How to act in case of accidents with chemicals or gases

If chemicals come in contact with parts of the body or the eyes

- Press the YELLOW CALAMITY button
- Remove yourself from the source of contamination.
- Take the **Diphoterine** sprayer which is in the vicinity of the workspace and immediately spray on the body part concerned. Do not use water.

In case of acids and alkalines

- Press the YELLOW CALAMITY button
- Remove yourself from the source of contamination.
- First completely use the **Diphoterine** sprayer, then continue the treatment with bags of **Diphoterine** from a green case labelled ACID / BASE Diphoterine.
- Open the white seal and pour the fluid on the concerned body part / eye without squeezing the bag. A bag becomes empty in approximately 3 minutes.
- In case of doubt, or when a large skin area is harmed also use a second bag.

In case of big quantities of chemicals

- Press the YELLOW CALAMITY button
- Remove yourself from the source of contamination.
- Take and use the green portable shower from the fire brigade hall, or instruct another person to do so.

In case of HF

- Press the YELLOW CALAMITY button
- Remove yourself from the source of contamination.
- Change from the Diphoterine sprayer to **Hexafluorine** as soon as possible.
- Get bags of **Hexafluorine** from a blue case labelled HF-KIT Hexafluorine.
- Open the white seal and pour the fluid on the concerned body part / eye without squeezing the bag. A bag becomes empty in approximately 3 minutes.
- In case of doubt, or when a large skin area is harmed also use a second bag.

In case of big quantities of HF-containing chemicals

- Press the YELLOW CALAMITY button
- Remove yourself from the source of contamination.
- Take and use the green portable shower from the fire brigade hall, or instruct another person to do so.

In case of organic fluids

- Press the YELLOW CALAMITY button
- Remove yourself from the source of contamination.
- Absorb the liquid with tissues, then thoroughly wash the concerned body parts with water and soap; let the tissues evaporate in the exhausted flow bench.
- If organic fluids are spilt in the eyes, rinse the eyes 10 to 15 minutes with an eye-shower.

In case of gases or chemical vapours

- Press the RED CALAMITY button
- Take the person immediately to a place with fresh air, regarding your own safety!
- If you consider the situation too dangerous then leave it to the Emergency Response Team.

Important contacts in case of questions or problems

The Emergency Response Team (Firefighter(FF), First Aid (FA))

Jan Warmerdam - First Commander (FF, FA)	Robert Verhoeven (FF, FA)
Koos van Hartingsveldt - Deputy Commander (FF,FA)	Martijn Tijssen (FF, FA)
Ron van Viersen - Deputy Commander (FF,FA)	Joeri Meijer (FF, FA)
Joost Berendse - Deputy Commander (FF,FA)	Mario Laros (FA)
	Henk van Zeijl (FA)
	Bianca Knot (FA)

The Safety Committee

Martijn Tijssen (Chair and Solar Cell Group)
Casper Juffermans (Director EKL)
Jan Warmerdam (Head Facility group)
Cassan Visser (CI100, SAL, MEMSlab)
Joeri Meijer (Mapper)
Tom de Weger (Occupational Health Service officer)

Radiological Health and Laser Safety officer

Koos van Hartingsveldt

Access and Visitor cards

Bianca Knot

Network Administrator

Wim Tiwon

Facility group

Check on sign under monitor in the entrance hall or near the CI 100 entrance which facility person is on call

Jan Warmerdam (Head facility group)	87745 / 0653133804
Robert Verhoeven	85960 / 0648875970
Ron van Viersen	86193 / 0634418115
Koos van Hartingsveldt	87382/ 0624373451
Maarten Witteveen	

- **PRESS RED BUTTON** in case of gasleak or other threat to the lab and its users
- **PRESS YELLOW BUTTON** if you need help from the Emergency Response Team immediately (eg chemical spill, lab evacuated, air flow low)
- **Facility emergency number 24/7** (only facility person on duty is called) **015-2510192** (ONLY FOR REAL EMERGENCIES that do not need a button pressed)
- **Report ALWAYS ALL** incidents and spills to the facility group.

In case of emergency outside office hours (all necessary emergency services are alarmed):

ALARM NUMBER: 112 fixed phone or **015-2781226** mobile phone